

Passed to a Friend, Not Freed

Auditing a Cooldown Timer on the Shared Piano-Practice-Room Booking Terminal

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The problem

A shared piano practice room turns over between players only if its last booker actually leaves the queue. Do rebooking cooldowns return a slot to circulation, or do they just change whose card is on file for it?

Research questions

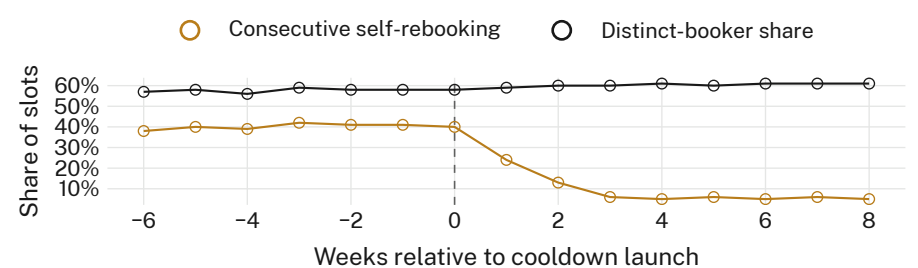
- Quantify consecutive same-card rebooking on the practice-room terminal before any cooldown.
- Test whether a mandatory cooldown shifts who books the room's next slot.
- Check whether cooldown-blocked bookers still recover the same slot by another route.

Study design

- $n = 6$ shared practice rooms · swipe-card terminal logs · 5-min sampling · 14-week window
- 45-min same-card cooldown live from 1 Sept 2025; enforced server-side against the terminal's last-booking log
- Booking pairs matched against a pre-registered "usual practice partner" field; threshold held constant, never adjusted post hoc

Findings

Consecutive same-card rebooking collapsed within three weeks of the cooldown; the room's distinct-booker share of those same slots barely moved over the following two months.



Consecutive same-card rebooking on the six monitored terminals fell from 41% to 6% of slots within three weeks of the 1 September 2025 cooldown; the distinct-booker share of those same slots rose from 58% to just 61% over the following eight weeks.

Interpretation & where next

The cooldown closed the loophole it targeted; the room's queue barely noticed. 88% of blocked rebooking attempts were followed, within the same slot boundary, by a card belonging to the blocked student's declared usual practice partner. Turnover reads as release only if the next card is checked against who last held the room, not just who is holding it. The pattern held across all six rooms, not only the busiest. Next: flag repeat partner hand-offs at the terminal before a slot is logged as freed.

Selected references

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De-identified card-pair logs only; no instrument, repertoire, or practice audio retained. Supported by a School of Continuous Improvement seed grant.

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